

Olivetti face

pixels: the convex blend is the output—you see a blur



No quantizer: the blend itself is the image—not a valid new face. So the clean output is a **verbatim stored image** (cosine 1.00).

RECALL

Protein sequence

a per-position $\arg\max$ quantizes the blend—you get a recombinant

SA-generated L F V G N L P D T T E E L K E F F S Q F G P V V S V

nearest natural L F I G G L A P Y T T E E N L K L F Y G Q W G K V V D V

teal = sites recombined away from the nearest natural (each is still a residue the family uses there)

every residue is one the family uses at that site (71/71, valid) • 38% of sites recombined (62% identity to nearest, novel) • passes RRM HMM

The convex blend lives in embedding space; $\arg\max$ snaps each of the 71 sites to one residue, so the *average is never seen*—different sites inherit from different neighbors.

NOVEL + VALID